

Sheet Neural

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```
[1]: import numpy as np
#Create a 1-dimensional NumPy array of length 7 with random integers between 0
    and 20.
x= np.random.randint(0,20,7)
print(x)
#Insert the value 100 at the third index of the array.
x=np.insert(x, 2, 100)
print(x)
#Then, delete the value at the fifth index of the array.
x=np.delete(x, 4)
print(x)
#Finally, calculate and print the minimum value and the unique values of the
    updated array.
unique=np.unique(x)
print(unique)
minimum=np.min(x)
print(minimum)
```

```
[18  7 13 17 16 19  8]
[ 18   7 100  13  17  16  19   8]
[ 18   7 100  13  16  19   8]
[  7   8  13  16  18  19 100]
7
```

```
[2]: # Create a 2D numpy array with dimensions (3,4) and fill it with random
    integers between 1 and 10
xy = np.array([[5,5,5,5],[4,3,3,3],[0,2,8,6]])
x0=np.array([[0,0,0,0],[0,0,0,0],[0,0,0,0]])
# Then, append a row of zeros at the end of the array, and reshape the array
    to (4,4).
X3 =np.append(xy,x0,axis=0)
#Finally, find the maximum value in the array.
maxvalue=np.max(X3)
print(maxvalue)
```

8

```
[3]: #Create a 2D numpy array with dimensions (4,5) and fill it with random integers
      ↳ between 1 and 100
arr= np.random.randint(100, size=(4, 5))

#Then, insert a column of ones at index 2, and delete the first row of the
↳ array.
arr=np.insert(arr,1,[1,1,1,1,1])
print(arr)
arr = np.delete(arr, 0, 0)
print("The array after Deletion:",arr)
#Finally, compute the mean of the remaining elements in the array.
mean = np.mean(arr)
print("The mean of the array:",mean)
```

```
[95  1  1  1  1  1 56 24 85 86 80 58 76 68 76 57  2 92 48 30 41 28 45 83
 27]
```

The array after Deletion: [1 1 1 1 1 56 24 85 86 80 58 76 68 76 57 2 92 48 30 41 28 45 83 27]

The mean of the array: 44.458333333333336

```
[5]: #Create a 3D numpy array of shape (2, 3, 4) filled with ones.
array= np.full(
    shape=(2,3,4),
    fill_value=1,
    dtype=np.int
)
print(array)
#Then, use the delete function to delete the second row of the second dimension
array = np.delete(array, 1,1)
print("The array After deletion:\n",array)
```

```
[[[1 1 1 1]
   [1 1 1 1]
   [1 1 1 1]]
```

```
[[[1 1 1 1]
   [1 1 1 1]
   [1 1 1 1]]]
```

The array After deletion:

```
[[[1 1 1 1]
   [1 1 1 1]]
```

```
[[[1 1 1 1]
   [1 1 1 1]]]
```

C:\Users\KIMO STORE\AppData\Local\Temp\ipykernel_6188\3196405470.py:5:

DeprecationWarning: `np.int` is a deprecated alias for the builtin `int`. To

silence this warning, use ``int`` by itself. Doing this will not modify any behavior and is safe. When replacing ``np.int``, you may wish to use e.g. ``np.int64`` or ``np.int32`` to specify the precision. If you wish to review your current use, check the release note link for additional information.

Deprecated in NumPy 1.20; for more details and guidance:

<https://numpy.org/devdocs/release/1.20.0-notes.html#deprecations>

`dtype=np.int`

```
[8]: #Create a 3-dimensional numpy array with shape (2, 3, 4) and fill it with
      ↪ random numbers between 0 and 1
      array= np.random.randint(2, size=(2,3, 4))
      print(array)
      #Then, reshape it into a 2-dimensional array with shape (6, 4).
      arr2 = np.reshape(array, (6, 4))
      print("The array After Reshaping:\n",arr2)
```

```
[[[1 1 1 1]
    [1 0 1 1]
    [0 0 0 1]]
```

```
[[1 0 1 0]
 [1 0 0 1]
 [1 0 0 1]]]
```

The array After Reshaping:

```
[[1 1 1 1]
 [1 0 1 1]
 [0 0 0 1]
 [1 0 1 0]
 [1 0 0 1]
 [1 0 0 1]]
```